Biospectra : ISSN: 0973-7057, Vol. 20(2), September, 2025, pp.111-116 ***Interdisciplinary Science***
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

The impact of Active Music Therapy (AMT) through Tabla playing on motor and cognitive skills in borderline intelligence: A case study

Karuna Nagarajan* & Sravan Kumar Vankeswaram
Centre for Distance Education, SVYASA Deemed to be University, Bengaluru, Karnataka, India.

Received : 11th April, 2025 ; Revised : 10th May, 2025
DOI:-https://doi.org/10.5281/zenodo.18531442

Abstract- Borderline intellectual functioning, defined by an IQ range of 71-84, falls just above the threshold for intellectual disability. Ayurveda emphasizes balancing *Vata, Pitta*, and *Kapha* for health. *Sadhaka Pitta*, a subtype of Pitta responsible for emotional and cognitive functions, impacts cognition, and its depletion may contribute to borderline intelligence. Ayurvedic practices, including warm food and percussion instrument use, are suggested to restore Pitta balance and enhance cognitive functions. To assess the effectiveness of playing Tabla as Active Music Therapy (AMT) for improving finger dexterity and cognition in an individual with borderline intelligence. A paired sample test design evaluated a 24-year-old male diagnosed with borderline intelligence before and after an 8-week Tabla intervention. Assessments included WAIS-IV for IQ, Bender's Gestalt Test for motor function, Warwick Edinburgh Mental Well-being Scale (WEMWS), Social Well-being Questionnaire, and Finger Dexterity Test. Verbal Comprehension (+7, $p < 0.05$) and Perceptual Reasoning (+8, $p < 0.05$) improved, increasing Full Scale IQ by 15 points. Visual-motor coordination errors decreased from 4 to 1. Finger dexterity, particularly in the left hand, progressed. Mental and social well-being scores showed improvement, though not statistically significant. Tabla playing as AMT enhanced verbal reasoning, spatial problem-solving, short-term memory, and finger dexterity in borderline intelligence. Longer interventions may provide broader cognitive and motor benefits.

Keywords: Sadhaka Pitta, Borderline Intelligence, Cognitive Function, Tabla, Percussion Instruments, Finger Dexterity, Active Music Therapy (AMT)

INTRODUCTION

Borderline intellectual functioning, defined by an IQ range of 71 to 84, exists just above the threshold for intellectual disability.[1] The condition arises from multiple factors, including genetic predispositions, environmental influences, prenatal and perinatal complications, and socioeconomic challenges.[2] Effective interventions involve early identification, specialized education, family support, vocational training, and therapeutic services aimed at enhancing cognitive and behavioral functioning.[3] Health in *Ayurveda* is the balance of *Vata* (Air), *Pitta* (Fire), and *Kapha* (Water), with Pitta governing heat, digestion, and metabolism. Formed by a combination of fire and water, *Pitta Dosha* governs all the processes related to metabolism and changes (mental and physical) that occur in the body.[4] *Sadhaka Pitta*, a subtype of *Pitta Dosha* located in the *Hridaya* (heart), plays a crucial role in cognitive functions and psychosomatic behaviour by actively preventing *Kapha* and *Tamoguna* (inertia) from slowing heart activity. Its imbalance can lead to emotional and psychosomatic disturbances, and its function aligns with neurotransmitters that facilitate communication between neurons and target tissues, impacting mental health and cognitive abilities[5] which includes Border line intelligence. Music, described

*Corresponding author :
Phone : 9740499958
E-mail : karuna.nagarajan@svyasa.edu.in

in *Brihatrayee* in *Ayurveda*, serves as a therapeutic tool for conditions like *Rajyakshma* (*Ojokshyaya*, an immunodeficiency disorder), aphrodisiac purposes, anxiety, and psychic disorders, helping restore focus and psychic strength in states of *Durmana (despair)* through music suited to individual preferences.[6]

Rhythm-based musical training has been shown to positively impact cognitive and emotional well-being across different populations.[7,8] Engaging in activities like drumming can strengthen executive functions (EF) and motor coordination by enhancing beat synchronization, which in turn supports self-regulation and EF development.[9] Drumming practice involves error monitoring and precise timing, helping improve attentional control and inhibitory functions, as reflected in enhanced performance on cognitive assessments such as the Integrated Visual and Auditory and Continuous Performance Test.[10,11] Additionally, these interventions contribute to the improvement of psychosocial and motor skills, particularly in children facing behavioural challenges.[12] To address the imbalance in *Sadhaka Pitta*, this case study explores the potential benefits of *Tabla* (an Indian percussion instrument) lessons as active music therapy (MT) to improve cognitive abilities and motor skills in an individual with borderline intellectual functioning. Active MT involves music-based interventions where patients engage interactively by playing or creating music.[13]

METHODOLOGY

This study explored the therapeutic potential of *Tabla* practice, a form of AMT, in improving cognitive abilities and motor skills in an individual with borderline intellectual functioning. The study employed a paired sample t-test design to assess changes across cognitive, physical, emotional, and social domains pre- and post-intervention. Data collection involved a combination of clinical diagnostics, standardized assessment tools, and interviews with the participant's family and music instructors to gain comprehensive insights.

Participant profile

Mr. X, a 24-year-old male, lives with his nuclear family and is the only child of a non-consanguineous marriage. A first-degree relative has a seizure disorder history.

Presenting Complaints

Persistent concerns include:

- Recurrent headaches with vomiting (11 years)
- Fatigue and lethargy (10 years)
- Communication difficulties (10 years)
- Academic struggles (5 years)
- Poor personal hygiene

No additional complaints reported. No additional associated complaints were reported.

Previous Treatment History

In 2017, Mr. X was diagnosed with borderline intelligence and low IQ at a renowned medical college.

Family and Developmental History

Mr. X was born full-term via LSCS, with normal birth weight and immediate crying. Childhood illnesses included recurrent headaches and vomiting. Developmental milestones were on track, and he maintains functional motor and speech abilities.

IQ Assessment

In 2017, his full-scale IQ was 83 ("Dull Normal"). Reading comprehension was below average, math skills mildly delayed. He shows independence in daily tasks but struggles with social interactions due to anxiety, introversion, and social communication disorder.

History of Present Illness

In 2024, Mr. X's mother reported longstanding health and academic issues. He has paused his degree for 1.5 years.

Study Population & Data Sources

This case study focuses on a single participant. Data were collected through interviews with family members, friends, and music instructor, alongside using validated assessment tools. Institutional ethical approval was obtained.

Intervention

The intervention involved structured *Tabla* lessons aimed at improving rhythm synchronization, motor coordination, and cognitive processing. A gradual progression of rhythmic patterns-*Thitta, Naa Thirakita*, and *Gi Gi Na Na Dha* - was employed to refine focus, timing, and fine motor skills. Sessions occurred twice a week for 30-40 minutes, supplemented by daily practice for 30 minutes between 7-8 pm, over 8 weeks.

Baseline assessments measured cognitive, emotional, physical, social, musical, and spiritual abilities to establish benchmarks. Tools included:

1. **WAIS-IV** for IQ and cognitive domains like verbal reasoning and memory.

2. **Bender Gestalt Test** for visual-motor integration.
3. **O'Connor Finger Dexterity Test** for coordination.
4. **Warwick Edinburgh Scale** for emotional well-being.
5. **Social Well-being Questionnaire** for interpersonal skill evaluation.

This method ensured consistent skill development and holistic improvements.

Assessment Tools

1. **Wechsler Adult Intelligence Scale (WAIS-IV):** Assessed cognitive domains such as verbal reasoning, memory, and processing speed.
2. **Bender Gestalt Test:** Evaluated fine motor skills and visual-motor integration.
3. **O'Connor Finger Dexterity Test:** Measured coordination and dexterity, particularly in the context of musical practice.
4. **Warwick Edinburgh Mental Well-being Scale (WEMWS):** Monitored emotional health and self-esteem throughout the study.
5. **Social Well-being Questionnaire:** Examined changes in interpersonal skills and **social engagement.**

RESULTS

The intervention, involving AMT through *Tabla* lessons, led to notable enhancements in cognitive functioning. Significant improvements were observed in Verbal Comprehension (+7, $p < 0.05$) and Perceptual Reasoning (+8, $p < 0.05$), culminating in a 15-point increase in Full Scale IQ. While Working Memory showed marginal progress (+1), Processing Speed exhibited a slight decline (-1). Errors on Bender's Gestalt Test decreased (from 4 to 1), reflecting enhanced visual-motor coordination shown in (Fig. 3 Pre – Post scores of various components of Wechsler's Intelligence Scale). Finger dexterity improved significantly in the left hand, although mixed outcomes were noted for the right as in (Fig. 1 Pre and Post Scores of O'Connor Finger Dexterity Test). Mental well-being, as in assessed as in (Fig. 2 Pre and Post scores of WEMWBS scale for Mental Well-being) using WEMWBS, increased by 8 points ($p = 0.0877$, non-significant), with positive trends in confidence, energy, and problem-solving, though relaxation and social connectedness saw a decline. Social well-being showed improvements in mental skills (+7) and vulnerability (+5), although these gains were not statistically significant ($p = 0.14$). Overall, AMT through *Tabla* lessons effectively boosted cognitive abilities and demonstrated promising trends in mental and social well-being, while highlighting areas for refinement in processing speed and social engagement.

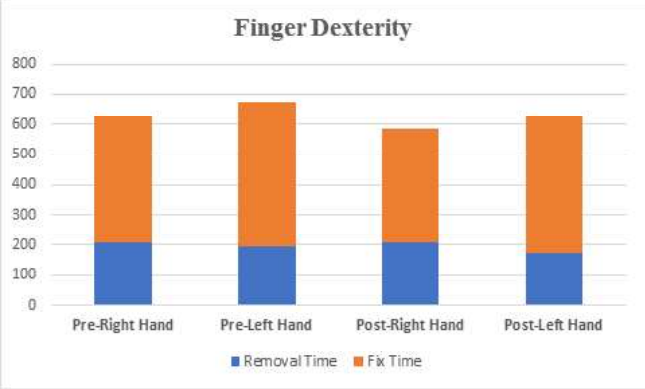


Fig. 1- Pre and Post Scores of O'Connor Finger Dexterity Test



Fig. 2- Pre and Post scores of WEMWBS scale for Mental Well being

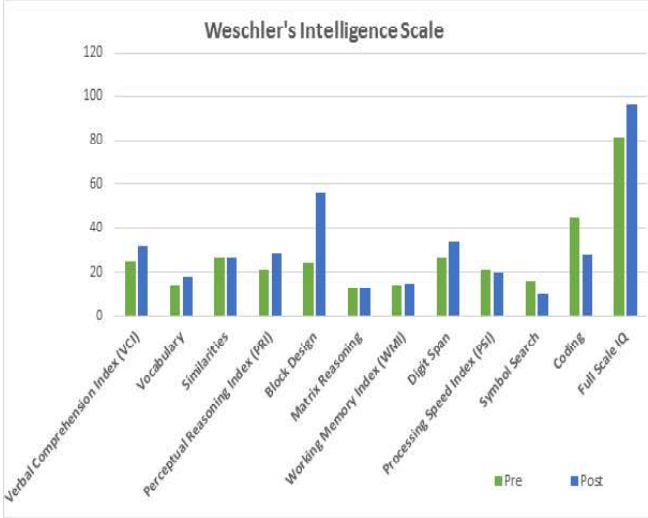


Fig.3- Pre -Post scores of various components of Wechsler's Intelligence Scale

DISCUSSION

Musical engagement has been widely recognized for its cognitive and emotional benefits, contributing to improved mental well-being, confidence, and cognitive clarity.[14]

The improvement of specific cognitive abilities can be linked to the coordinated finger and body movements required in AMT - *Tabla* playing. Research on rhythmic auditory stimulation-based movement training supports this, showing that synchronized auditory-motor engagement enhances cognitive functions, particularly in areas such as attention, memory, and executive functioning, even for individuals with impairments.[15] Social integration through music is another significant aspect of cognitive therapy. Percussion-based interventions refine social communication and engagement, although the effectiveness of individual versus group drumming in fostering social connectedness remains an area for further exploration.[16] From a neuropsychological standpoint, rhythmic training improves synaptic plasticity and cross-hemispheric brain connectivity, leading to measurable cognitive adaptability.[17] Indian classical music, known for its complex rhythmic structures, plays a crucial role in neurocognitive rehabilitation and cognitive stimulation.[18]

Motor skill development, particularly finger dexterity, is a well-established outcome of percussion training, aligning with research demonstrating that rhythmic exercises refine fine motor coordination and sensorimotor integration.[19] Studies on anticipatory control of fingertip forces suggest that structured drumming enhances precision and coordination,[20] while instrumental training such as *Tabla* practice strengthens neuromuscular control and dexterity.[21] The observed improvements in Verbal Comprehension and Perceptual Reasoning indicate that rhythmic engagement enhances higher-order cognitive processing, consistent with evidence that musical rhythm training improves attentional control and short-term memory.[22]

Furthermore, percussion-based interventions enhance synaptic connectivity and neural adaptability, fostering long-term cognitive benefits.[23] Musical engagement is also associated with psychological resilience, contributing to improved mood and cognitive flexibility.[24] Although the increase in WEMWBS scores was not statistically significant, the qualitative enhancements in confidence and cognitive clarity suggest potential long-term advantages.

Despite these promising findings, certain limitations should be acknowledged. The minimal changes in working memory and the slight decline in processing speed suggest the need for more structured tempo variations to optimize cognitive efficiency. Additionally, the lack of statistically significant improvements in social well-being highlights the need for further exploration into group-based drumming for enhanced social interaction.[16]

CONCLUSION

This study reinforces the cognitive, motor, and psychological benefits of rhythm-based interventions, particularly for individuals with borderline intelligence. Future research should explore the longitudinal effects of music therapy and individual variability in responsiveness to musical training.

FUNDING

The study has not received funding from any funding agency.

CONFLICT OF INTEREST

There are no conflicts of interest.

ACKNOWLEDGEMENTS

The authors gratefully acknowledge Dr. Mangesh Pandey for his valuable insights and support.

REFERENCES

1. **Martínez-Leal R, Salvador-Carulla L, Peltopuro M. 2020.** Borderline intellectual functioning: A poorly recognized phenomenon. *J Intellect Disabil Res*. **64(4):**287-298. https://doi.org/10.1111/jir.12712.
2. **Peltopuro M, Vesala HT, Ahonen T, Närhi V.M. 2023.** Borderline intellectual functioning and vulnerability in education, employment, and family. *Scand J. Disabil Res.* **25(1):**334-349. https://doi.org/10.16993/sjdr.965.
3. **Orío-Aparicio C., Bel-Fenellós C., López-Escribano C. 2025.** Understanding adaptive skills in borderline intellectual functioning: A systematic review. *Eur J Investig Health Psychol Educ.* **15(3):**40. https://doi.org/10.3390/ejihpe15030040.
4. **Savita D., Sharma A.K. 2021.** Concept of Tridosha Theory: A Critical Review. *Int J Creat Res Thoughts.* **9(1):**265-270.

5. **Jaiswal T., Sharma B., Nagpal S. 2024.** Sadhaka Pitta: The gatekeeper for mental health w.s.r. to neuro transmitter. *J. Ayurveda Integr Med Sci.* **9(4).** https://doi.org/10.21760/jaims.9.4.18.

6. **Upadhyay D., Mishra R. 2016.** Review of Music and its therapeutics with special reference to Ayurvedic Classics (Brihatrayee). *J. Ayurveda.* **2:**114-118.

7. **Bialystok E., Depape A.M. 2009.** Musical expertise, bilingualism, and executive functioning. *J Exp Psychol Hum Percept Perform.* **35(2):**565-574.

8. **Bugos J.A., Perlstein W.M., McCrae C.S., Brophy T.S., Bedenbaugh P.H. 2007.** Individualized piano instruction enhances executive functioning and working memory in older adults. *Aging Ment Health.* **11(4):**464-471.

9. **Williams K.E. 2018.** Moving to the beat: Using music, rhythm and movement to enhance self-regulation in early childhood classrooms. *Int J Early Child.* **50(1):**85-100.

10. **Sandford J, Turner A. 1994.** IVA+Plus: Interpretation Manual. Chesterfield (VA): BrainTrain;

11. **Lowry R.G., Hale B.J., Draper S.B., Smith M.S. 2018.** Rock drumming enhances motor and psychosocial skills of children with emotional and behavioral difficulties. *Int J Dev Disabil.* **65(3):**152-161.

12. **Draper, S., Lowry, R. G., Marino, L., Corrigan, R., Smith, M., & Williams, S. 2020.** The effects of a rock drumming intervention on children with additional educational needs. In: Dela F, Muller E, Tsolakidis E, editors. *European College of Sport Science Conference.* 236.

13. **McPherson T. 2019.** Active versus passive music therapy: A comparative analysis of their impact on patient outcomes. *J Music Ther.* **56(3):**327-345.

14. **Upadhyay A., Mittal S. 2019.** Exploring the relationship between music engagement and music function with emotion and psychological well-being. *Int J Indian Psychol.* **7(4):**125-140. https://doi.org/10.25215/0704.057.

15. **Wang, Y.N., Wen, X.N., Chen, Y., Xu, N., Zhang, J.H., Hou, X., Liu, J.P., Li, P., Chen, J.Y., Wang, J.H. and Sun, X.Y., 2024.** Effects of movement training based on rhythmic auditory stimulation in cognitive impairment: A meta-analysis of randomized controlled clinical trial. *Frontiers in Neuroscience,* **18:**1360935.

16. **Perkins, R., Ascenso, S., Atkins, L., Fancourt, D., & Williamon, A. 2016.** Making music for mental health: How group drumming mediates recovery. *Psychol Well-Being.* **6(1):**1 https://doi.org/10.1186/s13612-016-0048-0.

17. **Hanna-Pladdy B., Menken M. 2020.** Training-induced cognitive and neural changes in musicians: Implications for healthy aging. In: Cuddy LL, Belleville S, Moussard A, editors. Music and the Aging Brain. *Academic Press,* 221-243. https://doi.org/10.1016/B978-0-12-817422-7.00008-0.

18. **Upadhye S. M., Mahajan S. N. 2024.** The impact of body percussion on neuropsychological aspects with special reference to Tabla and Pakhawaj. *Sangeet Galaxy e-Journal.* **13(2):**25-32. Retrieved from www.sangeetgalaxy.co.in.

19. **Gzibovskis T., Marnauza M. 2012.** Development of young adults' fine motor skills when learning to play percussion instruments. *Music Educ Res.* **14(3):**365-380. Retrieved from ERIC.

20. **Venkadesan M., Valero-Cuevas F. J. 2009.** Anticipatory control of motion-to-force transitions with the fingertips adapts optimally to task difficulty. *J. Neurophysiol.* **102(5):**3012-3024. https://doi.org/10.1152/jn.00603.2009.

21. **Maqbool B., Rajaguru S. 2015.** Effect of Training in Tabla on the Development of Skills related to Motor and Neurological Functioning. *Int J Multidiscip Approach Stud.* **1(8):**95-102.

22. **Vega-Ávila, G.C., Afanador-Restrepo, D.F., Rivas-Campo, Y., García-Garro, P.A., Hita-Contreras, F., Carcelén-Fraile, M.D.C., Castellote-Caballero, Y. and Aibar-Almazan, A., 2022.** Rhythmic physical activity and global cognition in older adults with and without mild cognitive impairment: a systematic review. *International Journal of Environmental Research and Public Health,* **19(19):**12230.

23. Blum, Carlos. 2019. Cognitive Training Using Percussion Instruments for Persons with Mild Cognitive Impairment: A Feasibility Study. *J. Dement*, **2(1):** 108

24. Nijs L, Nicolaou G. 2021. Flourishing in resonance: Joint resilience building through music and motion. *Front Psychol.* **12:**1-11. https://doi.org/10.3389/fpsyg.2021.666702.

***